

CHE 322 THERMODYNAMICS (Unique Number 16480)
Course Syllabus

DESCRIPTION: This course covers the basic concepts of thermodynamics. Besides the first law (which was covered in CHE 317, a pre-requisite for this course), this includes the 2nd law of thermodynamics, and the criteria for phase equilibrium, which is the basis of most separation problems. We will discuss models to describe the nonideality of gas and liquid phases and use these models to solve vapor/liquid, liquid/liquid, gas/liquid and solid/liquid phase equilibrium problems.

GOALS: The ability to solve problems that require knowledge of both the 1st and 2nd laws, including practical problems involving power cycles and refrigeration cycles, is a fundamental skill required of all chemical engineers. In addition, chemical engineers need to be able to calculate the phase equilibrium of both ideal and nonideal gas and liquid mixtures and, subsequently, to be able to design separation processes based on those phase splits. The ability to design separation processes is one of the major things that differentiates chemical engineers from other types of engineers and chemists. By the end of the semester you should be able to:

- * evaluate the feasibility of a process
- * analyze power and refrigeration cycles
- * determine the compositions in gas, liquid and solid phases of all types of ideal and nonideal mixtures that are in equilibrium.
- * make judgments on which models would be most appropriate for a particular system
- * determine the validity and correctness of AI generated responses to chemical thermodynamics queries

REQUIRED TEXT: *Introductory Chemical Engineering Thermodynamics*, 2nd Ed., J. Richard Elliott and Carl T. Lira, 2012, Prentice-Hall.

CLASS: TTh 11:00-12:30 CPE 2.208 In-person only; attendance required
 W 4:00-6:00 GAR 0.102 In-person only; note that attendance at these recitation sections is **required**. The regular exams and most of the weekly recitation quizzes will take place during these sessions.

INSTRUCTOR: Professor Joan F. Brennecke email: jfb@che.utexas.edu EER 3.614

TA: Jun Park email: junwon.park@austin.utexas.edu EER 3.612

GRADERS: Rohan Bhavsar
 Paul Gonzalez

TUTOR: Joey Jabbour

OFFICE HRS: The Tutor, TA and Dr. Brennecke will hold regular office hours. In addition, you are welcome to contact Dr. Brennecke by email to make an appointment at another time. Office hours will start Tuesday, Jan. 20.

Dr. Brennecke	Wednesdays 10 am - noon	EER 4.702
Jun Park	Mondays 2-3 pm	EER 4.702
	Tuesdays 2-3 pm	EER 4.702
Joseph Jabbour	Mondays 6-8 pm	CPE 2.222
	Tuesdays 4-6 pm	EER 4.704
	Fridays 2-3 pm	CPE 2.222

GRADES: Regular and final exams (see below) 70

Recitation quizzes	25
Lecture quizzes	5

The full range of available grades (including + and -) will be utilized.

REGULAR EXAMS: The tentative dates of the three regular exams are: **February 11, March 11 and April 15**. They will be held during the scheduled recitation sections on Wednesdays from 4-6 pm in GAR 0.102. The exams will be closed book, closed classnotes. Please ensure that you have a working non-transmitting device calculator to bring to the exams (and quizzes). Necessary data, model equations or formulas will be provided. However, you may bring one 8 ½" x 11" sheet with notes on both sides to the first exam, two sheets to the second exam, three sheets to the third exam, and four sheets to the final.

FINAL EXAM: The final exam is scheduled for **Saturday, May 2** from 8-10 am. The final is cumulative, but emphasis will be placed on material since the third exam. Like the regular exams, the final is closed book/closed notes. As indicated above, you may bring four 8 ½" x 11" sheet with notes on both sides to the final.

To provide extra flexibility, the lowest absolute score of your four exam grades will only count half as much as the other three exams. So your three exams with the highest absolute scores will each count 20% of the total points for the course and your exam with the lowest absolute score will count 10%.

I must receive a notification from Student Emergency Services (<https://deanofstudents.utexas.edu/emergency/>), before the missed class in all but the most dire circumstances, in order to reschedule an exam.

HOMEWORK and RECITATION QUIZZES: Homework will be assigned for nearly every week during the semester. Where there is a simple numerical answer, those answers will be provided with the assignment. Homeworks will not be collected. However, you will have a closed book/closed note quiz each week (usually during the Wednesday recitation session). The recitation quizzes will be composed of problems very similar to homework problems, generally with some simple modifications. It may also include concept questions similar to the things we will discuss in the TTH lectures. Please ensure that you have a working non-transmitting device calculator to bring to the quizzes. Necessary data, model equations or formulas will be provided. The 8 ½" x 11" sheets are only for exams; you cannot bring any notes to the weekly recitation quizzes. The recitation quizzes will generally be 30 minutes to 1 hr. After the quizzes are collected, Jun or Dr. Brennecke will answer any questions on that week's homework and quiz, and have a few discussions on the use of AI for learning chemical thermodynamics. One of the recitation quizzes (Quiz #11) will require individual demonstration of proficiency in the use and application of a computer-based modeling program. This will require signing up for short individual meetings with Dr. Brennecke at times outside of class roughly in the Week 11-13 timeframe. Another of the "quizzes" (Quiz #0) will also be a short individual meeting with Dr. Brennecke in the first two weeks of class to assess your recollection of energy balances. The score on Quiz #0 will be either 50/50 (if you show up for your appointment) or 0/50 (if you don't show up). In order to provide you with flexibility, I will not count your lowest recitation quiz score. Recitation quizzes cannot be made up. A solution key to the homework problems and the recitation quiz will be posted online after each recitation quiz is completed. If I receive a notification of an excused absence from Student Emergency Services (<https://deanofstudents.utexas.edu/emergency/>) before (in all but the most dire circumstances) a missed quiz, then that quiz will also be dropped. Recitation quizzes will count for 25% of your total grade in the course.

You may work together with your classmates on homework. However, please do not deny yourself the opportunity to struggle with difficult concepts and problems to try to figure them out, before seeking assistance from others. This is an important part of learning. It is a truly exceptional individual who can solve chemical engineering problems without practicing. Homework problems is the way that you practice how to solve problems.

LECTURE QUIZZES: We will have very short closed book/closed notes quizzes, primarily on concepts or very simple calculations, in the TTh lectures. Therefore, please bring a non-transmitting device calculator to every class. The lecture quizzes will count for a total of 5% of your grade in the course. Since there will be ~20 of these quizzes throughout the semester amounting to just 5% of your grade, there will not be any “dropped” or excused absences from lecture quizzes and there will not be any ‘make-up’ lecture quizzes.

REGRADE POLICY: The policy on exam and recitation quiz regrades is as follows: exam and quiz solutions will be gone over in class/recitation section or posted on Canvas when the graded exams are returned and graded recitation quizzes posted. If you feel that your exam or quiz has not been graded correctly after reviewing the solutions, you must hand back the exam/printed out quiz with a written explanation of the oversight by the beginning of the next Tuesday or Thursday class period after the graded exam is returned to the class or graded quiz posted. We have the prerogative to regrade the entire exam/quiz, so that your final grade may be corrected up or down depending on the regrade of the entire exam/quiz. If you are not in class when the graded exam is returned, then it is your responsibility to collect your graded exam from Dr. Brennecke/Jun prior to the next TTh class. Otherwise, you forfeit your option to request a regrade, if one is necessary.

READING ASSIGNMENTS: Generally, I will tell you the sections in the textbook that correspond to the material that will be covered in class. I expect that you will be prepared for each class. This means that you have read the sections in the textbook before class and that you are ready to discuss the topic being covered that day. You are accountable for all the material assigned to you, but only a portion of this material will be explicitly covered in class. My philosophy is that class time should be used to help you understand difficult concepts, to develop problem solving skills and to emphasize points not made in the textbook. I will try not to use class time to go over material you can easily learn through self-study with the textbook.

HELP SESSIONS: Help sessions may be scheduled periodically outside of class throughout the semester to go through additional sample problems and answer questions before exams.

ACADEMIC INTEGRITY: The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the University is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community. As a student of The University of Texas at Austin, you have agreed to abide by the core values of the University and uphold academic integrity. Academic dishonesty will not be tolerated in CHE 322. Forms of academic misconduct include (but are not limited to) plagiarism, use of unauthorized materials for exams or quizzes, and changing solutions to returned exams. Further information is available at: <https://deanofstudents.utexas.edu/conduct/avoiding-academic-misconduct.php>

The work you turn in must be your own. Exams and quizzes are to be done only by you. All transmitting devices (e.g., cellphones) must be packed away during exams and quizzes. Your calculator must be a non-transmitting device. Only one student can leave the classroom at a time during an exam or quiz. Communicating the lecture quiz question(s) or answer(s) to anyone, including someone who is not in the classroom, is considered academic dishonesty. If you have questions about anything that you are not sure of, come speak to Dr. Brennecke.

STUDENTS WITH DISABILITIES: If you are a student with a disability, or think you may have a disability, and need accommodations please contact Disability and Access (D&A). You may refer to D&A’s website for contact and more information: <https://disability.utexas.edu/> If you are already registered with D&A, please deliver your Accommodation Letter to me as early as possible in the semester so we can discuss your approved accommodations.

COURSE OUTLINE

Chapter		approximate # of hours of lecture
	Introduction, organization	1/2
2	1 st law review	2 1/2
17	Reaction equilibrium	3
4	Entropy	4
5	Applications (liquifaction, power cycles, refrigeration cycles)	5
6	Thermo properties of real substances	3
7	Equations of state	3
8	Departure functions	2
9	Equilibrium for pure substances	4
10	Introduction to multicomponent mixtures and Raoult's Law	3
11, 12, 13	Activity models and vapor-liquid equilibria	6
14	Liquid-liquid phase equilibria	4
14	Solid-liquid equilibria	1

SCHEDULE

Week 1	Jan. 13 and 15	Lectures
	Jan. 14	Review of use of steam tables
Week 2	Jan. 20 and 22	Lectures
	Jan. 21	Quiz 1 over HW#1
Week 3	Jan. 27 and 29	Lectures
	Jan. 28	Quiz 2 over HW#2
Week 4	Feb. 3 and 4	Lectures
	Feb. 5	Quiz 3 over HW#3
Week 5	Feb. 10 and 12	Lectures
	Feb. 11	Exam #1
Week 6	Feb. 17 and 19	Lectures
	Feb. 18	Quiz 4 over HW#4
Week 7	Feb. 24 and 26	Lectures
	Feb. 25	Quiz 5 over HW#5
Week 8	Mar. 3 and 5	Lectures
	Mar. 4	Quiz 6 over HW#6
Week 9	Mar. 10 and 12	Lectures
	Mar. 11	Exam #2
Week 10	Mar. 24 and 26	Lectures
	Mar. 25	Quiz 7 over HW#7
Week 11	Mar. 31 and Apr. 2	Lectures
	Apr. 1	Quiz 8 over HW#8
Week 12	Apr. 7 and 9	Lectures
	Apr. 8	Quiz 9 over HW#9
Week 13	Apr. 14 and 16	Lectures
	Apr. 15	Exam #3
Week 14	Apr. 21 and 23	Lectures
	Apr. 22	Quiz 10 over HW#10

Two of the recitation quizzes (Quiz #0 and Quiz #11) will require signing up for short individual meetings with Dr. Brennecke at times outside of class during the first two weeks of class and roughly in the Week 11-13 timeframe.